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Nutrient-Use Efficiency and Productivity of Tomato Are Affected by Magnesium Ammonium Phosphate Fertilization Derived from Lignocellulosic Biomass

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ABSTRACT: Conventional fertilizers release nutrients quickly thereby resulting in a poor nutrient use efficiency and environmental losses. In the present study, we developed a MAP fertilizer from lignocellulosic biomass and compared nitrogen phosphorus (NP) release characteristic from it with those of urea and DAP. The nutrient release studies showed that N from MAP was released 9 folds slower than from urea and 3-folds less than DAP. Phosphorus release was also found to be 2.4 times lower compared to DAP, suggesting controlled (and sustained) nutrient availability.

The tomato field test (*Solanum lycopersicum* L.) showed that MAP treatment significantly enhanced plant vigor and sustained leaf greenness throughout the growing season. When compared with control application of fertilizers, MAP treated plants enhanced fruit yield/plant by 64%, Fruit weight by 63 % and wet and dry biomass respectively in the order of 21% and 23%. Fruit yield, fruit weight and wet and dry biomass were all increased significantly in MAP treated plants compared with untreated and control treatments. These results indicate that lignocellulosic biomass-derived MAP fertilizers not only synchronize nutrient supply with crop demand, but also enhance N and P use efficiency substantially in addition to the increase in tomato productivity. Their utility as environmentally friendly alternatives to mineral fertilizers in intensive horticultural systems is stressed.

1. INTRODUCTION

The world population is projected to reach 9.7 billion by 2064 [1], and, hence, there will be an enormous demand on agricultural systems to increase the production of food. Under such scenarios, current fertilization strategies seem insufficient to meet the future demand for food unless resources are used more effectively. But the nutrient use efficiency (NUE) of traditional fertilizers is intrinsically low, with 30-35% applied nitrogen (N), 18-20% phosphate (P), and 35-40% potassium being efficiently utilized by crops [2]. As a result, farmers apply excessive fertilizers to support yields, resulting in higher production cost and more energy consumption. This method has subjected farming systems not only to economic/market limitations but it also questions their sustainability from an agricultural perspective. Over application of fertilizer has been broadly associated with environmental problems such as greenhouse gas emissions, nutrient leaching, eutrophication of water bodies, and soil degradation [3].

Nitrogen is a basic element for plant growth, yield formation and product quality [4] and phosphorus influences root development, energy transfer in the form of ATP and reproductive processes. Magnesium, as the central atom of chlorophyll is necessary for photosynthesis, carbon metabolism and enzymatic regulation. Huge nitrogen and phosphorus fertilizers quantities are routinely used to maintain high yields in intensive cropping system. Nevertheless, NUE is continuously decreasing, especially in developing agricultural areas [5,6]. The inefficacy of traditional fertilizers is mainly due to the rapid solubility and

leaching, volatilization, surface runoff and fixation reactions in soils as an aftermath nutrient losses. These restrictions have redirected attention toward the advancement of new fertilizer technologies that can enhance nutrient use efficiency and synchronize nutrient availability with crop demand.

Controlled-release and slow-release fertilizers (CRFs/SRFs) as an effective supplements or substitutes for conventional fertilization by controlling the release of nutrients from either physical coating or chemical modification thus reducing loss and increasing NUE [7-14]. In recent years, the biodegradable and renewable nature of carrier materials has gained more attention especially carriers produced from lignocellulosic biomass such as agricultural residues and agro-industrial by-products. However, lignocellulosic substrates are characterized by favoured properties, such as high porosity, rich surface functional groups, renewability and biodegradability they can be used for the matrix of nutrient encapsulation loading and sustained release. The imbedding of key nutrients (N, P, Mg and others) as components in Lignocellosic structures during expanding growth provides for a diffusion controlled release of these nutrients to the ecosystem and also input large amounts of dissolved organic C into soil systems [15].

In comparison with traditional polymer-coated fertilizers, lignocellulosic biomass-based controlled-release fertilizers are environmentally friendly and have reduced reliance on petroleum products and good effects on the soil. The porous structure in the lignocellulosic carriers also plays a role in increasing adsorption and retention of nutrients, thereby prolonging nutrient availability in the rhizosphere with reduced leaching losses [16]. In addition, the gradual decomposition of these carriers triggers soil microbial activity, enhances soil physical characteristics, and promotes nutrient cycling so that fertilizer application approaches sustainable and circular agriculture [17,18].

According to biomass-based materials, biochar is a carbonaceous material generated from pyrolysis of biomass resources and enriched with high surface area and powerful nutrient and water preservation ability [19,20], which has been used widely in fertilizer products. Despite the agronomic and environmental advantages of biochar-based fertilizers, their long-term safety and sustainability are still controversial [21]. Pyrolysis may generate potentially hazardous organic pollutants such as Polycyclic Aromatic Hydrocarbons (PAHs), dioxins, and furans [22-29]. Moreover, biochar has the potential to adsorb and accumulate heavy metals (e.g., Cu, Cd, and As), which may be a long-term risk for soil quality, crop health, and food safety [30-34]. These environmental problems emphasize the importance of developing environmentally friendly alternatives systems of lignocellulosic biomass as a source for fertilizers, which can also provide slow release without causing secondary risks to the environment.

Tomato (*Solanum lycopersicum* L.) is one of the most economically important and commonly grown horticultural crops in all regions of the world [35], appreciated for its nutritional contribution to human diets [36]. Balanced nutrient supply, especially nitrogen, phosphorus and magnesium are the key factors to tomato yield and quality in terms of vegetative growth regulation, photosynthetic capacity and fruit development. Although the control- and slow-release fertilizers of many crop species were investigated extensively, there has been little interest in lignocellulosic biomass-based slow-release fertilizers containing nitrogen, phosphorus, and magnesium applied to tomato cropping systems. Compared with traditional fertilizers, these bio-based formulations are biodegradable and environmentally friendly, and they may have the advantage of increasing nutrient use efficiency via the maintenance of soil quality.

Thus, the present study was conducted to evaluate the efficacy of lignocellulosic biomass based nitrogen/phosphorus/magnesium-enriched slow-release fertilizers on performance response of tomato in terms of growth, yield, fruit quality and nutrient use efficiency. These results should be important in providing environmentally friendly fertilizer technology that can improve tomato production and reduce the environmental impacts derived from conventional fertilization practices.

EXPERIMENTAL SECTION

Materials

Laboratory grade reagents, orthophosphoric acid ($\geq 85\%$, Emplura, Merck), magnesium oxide (Emplura, Merck) and 25% aqueous ammonia solution (Emplura, Merck), were commercially obtained and used as received. The lignocellulosic materials which were used in the development of the slow release fertilizer matrix are comprised approximately by 46% cellulose, and 25% lignin and 23% hemicellulose. Commercial DAP and urea were purchased from a local dealer and used as standard fertilizers in comparative trials.

Preparation of Lignocellulosic biomass based MAP fertilizer:

Take 1.0g finely ground lignocellulosic biomass (particle size $\sim 0.2\text{mm}$) was weighed in to a glass beaker. Then, 0.56mol of orthophosphoric acid (calculated based on initial biomass content) was added to the biomass, and the suspension was stirred at 150rpm and kept for 2h at 70°C in order to allow impregnation with acid and partial hydrolysis of the biomass. Thereafter, 0.2g of magnesium oxide was added into the reaction medium and stirring was prolonged under similar conditions for further 1hour. The resultant solution was slowly neutralized to approximate pH 7 at room temperature while under constant hydrodynamic stirring by addition of 25% aqueous ammonia. To control the exothermic neutralization, during ammonia addition the beaker was kept at an ice bath. The neutralized suspension was heated at 90°C in a water bath for 2h to induce precipitation of the magnesium ammonium phosphate compound. The resultant solution was then evaporated under vacuum on a rotary evaporator. The end product a lignocellulosic magnesium ammonium phosphate (MAP) also analyzed and characterized.

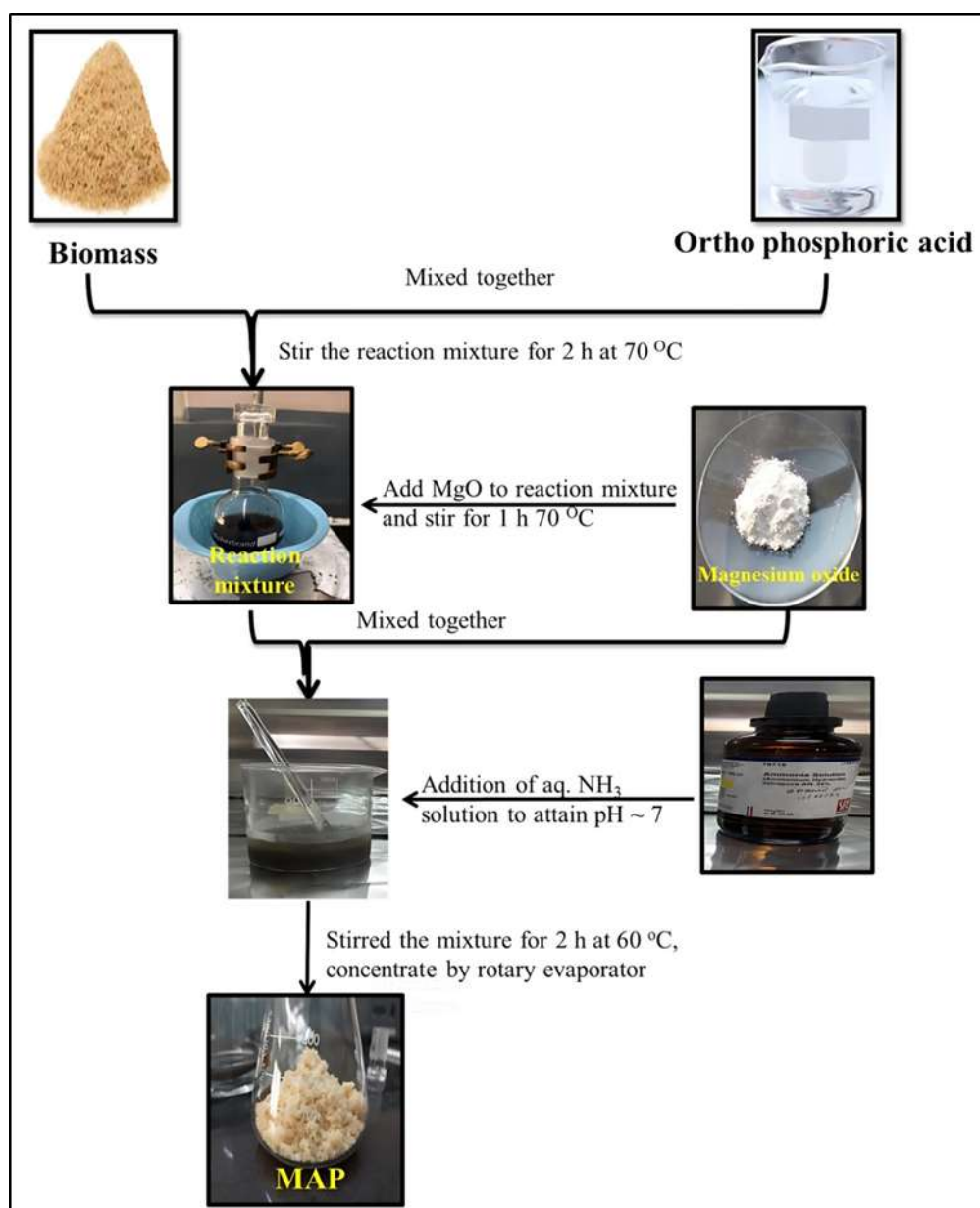


Fig. 1. Synthesis procedure of lignocellulosic biomass-based MAP sustainable-release fertilizer.

MATERIALS CHARACTERIZATION

X-ray Diffraction (XRD)

The X-ray powder diffraction (XRD) pattern of magnesium ammonium phosphate (MAP) was obtained on a D2 PHASER diffractometer (Bruker AXS) in the 2θ range from 10° to 80° at a scan rate of $5^\circ/\text{min}$. For analysis, the powdered samples were kept over a hot air oven at 105°C for 3 h to dry and maintain sample stability.

Scanning Electron microscopy (SEM):

The surface morphology of MAP was observed with a JEOL JSM-6490A scanning electron microscope (SEM) at 20.0 kV accelerating voltage. The microstructure and surface texture of the core were observed using SEM images at different magnifications, such as 15,000X and 60,000X.

Fourier Transform Infrared (FTIR) spectroscopy:

Functional groups of magnesium ammonium phosphate (MAP) were determined on PerkinElmer FTIR spectrometer with attenuated total reflectance (ATR) as the installed accessory. Spectrum was obtained for the wavenumber range $4000\text{--}400\text{ cm}^{-1}$ keeping an adequate resolution to detect the characteristic vibrational modes.

Thermogravimetric analysis (TGA):

Thermal stability of MAP was tested by a Q500 thermo gravimetric analyzer (TA Instruments). The sample was heated from room temperature to the desired range at $10^\circ\text{C}/\text{min}$ under nitrogen with a flow rate of $50\text{ mL}/\text{min}$. The thermogram obtained was applied for the evaluation of decomposition profile and thermal behaviour of the materials.

Analytical characterization of energy crop based lignocellulosic magnesium ammonium phosphate (MAP) fertilizer:

A range of established and validated analytical techniques were utilized to quantify the nutrient composition and elemental analysis of the lignocellulosic magnesium ammonium phosphate (MAP) fertilizer, which was synthesized. These studies involved the determination of total nitrogen, available phosphate, magnesium contents and organic carbon by standard procedures. Details of the specific analytical processes for each parameter are provided as follows.

Total Nitrogen by Kjeldahl method:

The total nitrogen in the lignocellulosic magnesium ammonium phosphate (MAP) fertilizer sample was analyzed by classical Kjeldahl method including digestion, distillation and titration. Digestion of about 0.5g of the sample took place in concentrated H_2SO_4 and catalysts such as potassium sulfate/copper or selenium. This process changes organic nitrogen to ammonium sulfate. Upon complete digestion the mixture was made alkaline by the addition of excess sodium hydroxide (NaOH) and ammonia (NH_3) was evolved. The free ammonia was distilled and collected in a graded boric acid solution, which was titrimetered with standardized sulfuric acid. The total nitrogen was determined according to the volume of titrant consumed [37].

Total Phosphate with the Vanadate-Molybdate Photometric Method

The phosphate extracted from lignocellulosic MAP fertilizer magnesium ammonium phosphates (MAP) was quantified by vanadate-molybdate colorimetric method according to APHA standard method, 4500-P C (APHA, 1998), in which the available orthophosphate or phosphate in this fertilizer reacts with ammonium molybdate and ammonium vanadate under acidic conditions of sulphate-vanadomolybdophosphoric acid complex giving a yellow complex. The generated chromophore showed a absorbance maximum at 470 nm. Absorbance was measured by a UV-Vis spectrophotometer against the reagent blank, and phosphate concentration was calculated using a standard calibration curve established with known concentrations of phosphate [37].

Total Magnesium by complexometric titration with EDTA

The combined magnesium content in lignocellulosic magnesium ammonium phosphate (MAP) fertilizer was determined by complexometric titration with the chelating agent ethylenediaminetetraacetic acid (EDTA). A known amount of the sample solution containing magnesium ions pH adjusted to 10 with an ammonia–ammonium chloride buffer. Eriochrome Black T was the metal ion indicator that gives a wine red complex with divalent metal ions. EDTA selectively chelates the magnesium

ions during titration, releasing the indicator and producing a clear color change from wine red to blue at the endpoint. Magnesium content was determined by the volume of standardized EDTA solution which had to be added until the colour of indicator changed [37].

Organic Carbon by modified Walkley–Black method

Lignocellulosic magnesium ammonium phosphate (MAP) fertilizer were analyzed for organic carbon using modified Walkley-Black wet oxidation method. Weigh a known amount of the sample and heat with excess potassium dichromate ($K_2Cr_2O_7$) and concentrated sulfuric acid (H_2SO_4) to oxidize organic material. The heat of acid dilution acted to promote oxidation. The excess unreacted dichromate was then titrated with standard ferrous ammonium sulphate (FAS) solution using a Ferroin indicator. Organic carbon was determined by the reduction of dichromate during oxidation [38].

The sample analyzed by the above techniques and results obtained were as follows.

Nutrient composition in MAP	
Nitrogen (% w/w)	18.0
Phosphorus as P_2O_5 (% w/w)	46.1
Magnesium (% w/w)	0.1
Organic carbon (% w/w)	0.56

Table-1: Nutrients content of lignocellulosic biomass-based Magnesium ammonium phosphate (MAP) sustained release fertilizer.

RESULTS AND DISCUSSION

X-ray diffraction (XRD) The X-ray diffraction (XRD) pattern of the prepared Lignocellulosic magnesium ammonium phosphate (MAP) composite is illustrated in Fig. 2. The major phase detected is Magnesium ammonium phosphate ($MgNH_4PO_4$) with characteristic reflexions at $2\theta \sim 16^\circ$, 23° and 29° and which correspond to the monoclinic structures previously reported in literature [39,40]. Other peaks at $2\theta = 34^\circ$, 36° and 49° can be assigned to monoammonium phosphate ($NH_4H_2PO_4$), which appear on the diffractograms of composites (41,42) this indicates the existence of MAP in amorphous lignocellulosic biomass. This phase mixture shows successful inclusion of magnesium and phosphate in composite structure and notably without an amorphous contaminant from the lignocellulosic hydrogel (which is used as a support matrix, not to integrate an amorphous component). This type of biomass which consisted mainly of cellulose, hemicellulose and lignin lacks strong diffraction peaks but offers an excellent support for nutrient encapsulation and release. Research articles on phosphorylated and mineral loaded cellulose-based materials revealed that these structures assist in holding nutrients to prevent leaching and allow controlled release of the same into soil matrix [43, 44].

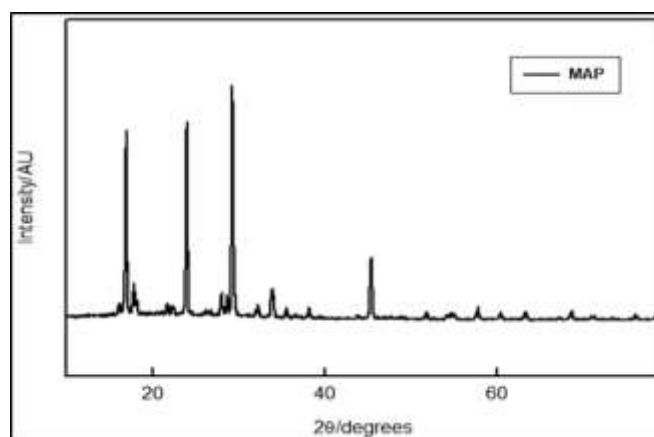


Fig. 2. X-ray diffraction of lignocellulosic biomass-based MAP sustained-release fertilizer.

Scanning electron microscopy (SEM)

The energy dispersive X-ray spectroscopy (EDS) elemental mapping image superimposed on the scanning electron microscopy (SEM) micrograph gives the spatial distribution of important nutrient elements namely, magnesium (Mg), nitrogen (N) and phosphorus); in this lignocellulosic matrix. The gray scale background also shows the cuticle or rather morphology of biomass surface, comprising uneven smoothen surfaces granules resulting likely from partially decomposed plant cell wall elements [45].

Red color of Mg elemental localized in red represents magnesium known as the main composition of the MAP compound. Nitrogen is hit in the yellow (probably of ammonium ion). The color turquoise is that of phosphorus, one of the underlying foods. The overlay reveals homogenous as well as dispersed location of all three elements on the biomass, indicating that the impregnation of MAP into the lignocellulosic material was successful. Notably, the phosphorus and magnesium signals are highest at the surface protrusions, indicative of preferred crystallization or surface complexation sites, whereas nitrogen is distributed in an extended manner which may implicate integration into both MAP phase and organic matrix. The lack of noticeable element clustering or segregation supports the homogeneous preparation of the composite that is also essential for controlled nutrient delivery. This is because it confirms the hypothesis of efficient nutrient loading and strong binding between MAP and the functional groups (OH, COO⁻) on lignocellulosic according to spatial homogeneity. These results give support to the viability of using lignocellulosic biomass as a nutrient carrier for its use in sustainable slow release fertilizers [46-48].

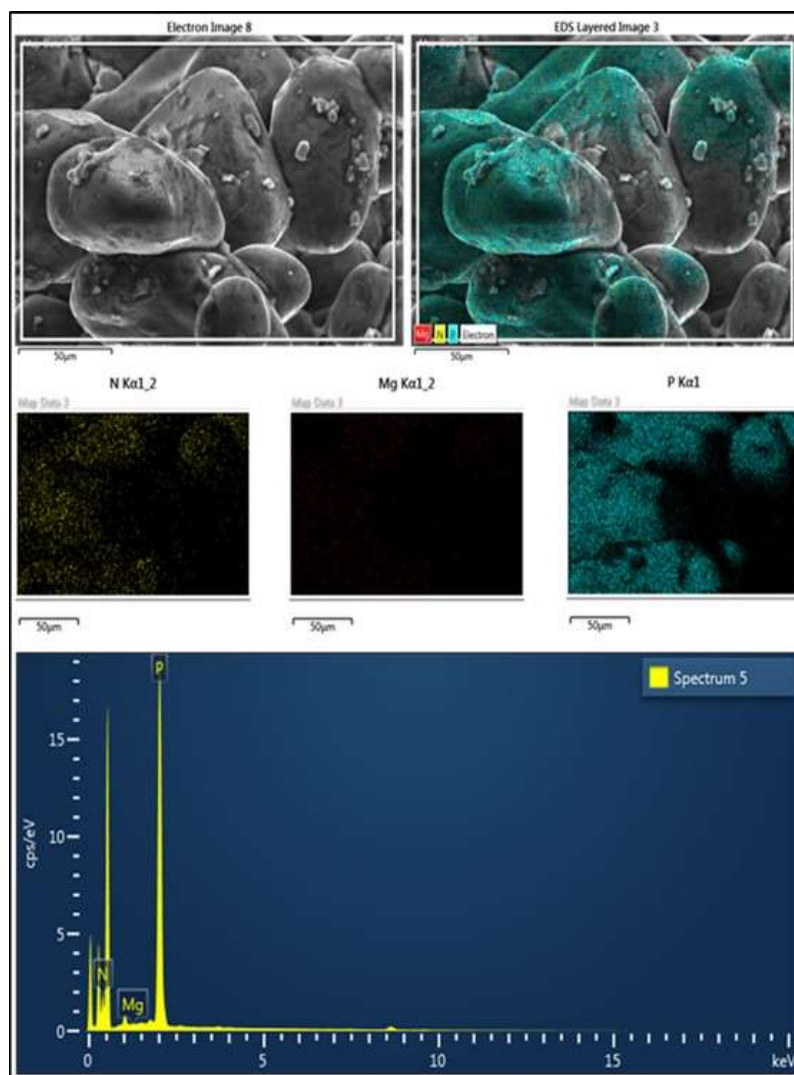


Fig. 3. Energy-dispersive X-ray spectroscopy (EDS or EDX) spectrum of the lignocellulosic biomass derived magnesium ammonium phosphate (MAP) controlled-release fertilizer.

The FT-IR spectrum of the LCB-based MAP fertilizer is depicted in Fig. 4. The vibration frequencies attributed to the modes of NH_4^+ , PO_4^{3-} fundamental vibrating units belong to Mg-O bond relations and taken from the literature [49, 50]. In the Lignocellulosic Biomass-derived Magnesium Ammonium Phosphate (MAP) spectrum stretching vibrations of O-H and N-H groups of MAP appears at 3216 & 3039 cm^{-1} [44], whereas asymmetric bending ($\delta_{\text{as}}\text{NH}_4$) are found at 1434 cm^{-1} . The appearance of new peaks (2322 cm^{-1} P-H bond), 1258 cm^{-1} (P=O bonds) and a shoulder at 970 cm^{-1} , due to P-OH group as well as at 825 cm^{-1} can be attributed to P-O-C stretching in phosphate ester [50,51] Shows the success reaction of phosphorylation. Furthermore, two bands at approximately 2833 cm^{-1} and 3039 cm^{-1} are assigned to a second overtone band and the combination frequency of asymmetric bending ($2\delta_{\text{as}}\text{NH}_4$) vibrations, respectively [43]. The vibrational bands of PO_4^{3-} ion attributable to the $\nu_2(\text{PO}_4^{3-})$, $\nu_4(\text{PO}_4^{3-})$, $\nu_1(\text{PO}_4^{3-})$ and $\nu_3(\text{PO}_4^{3-})$ vibrational modes were found to manifest respectively in the 537 cm^{-1} , 722 cm^{-1} , 847 cm^{-1} and 1045 cm^{-1} .

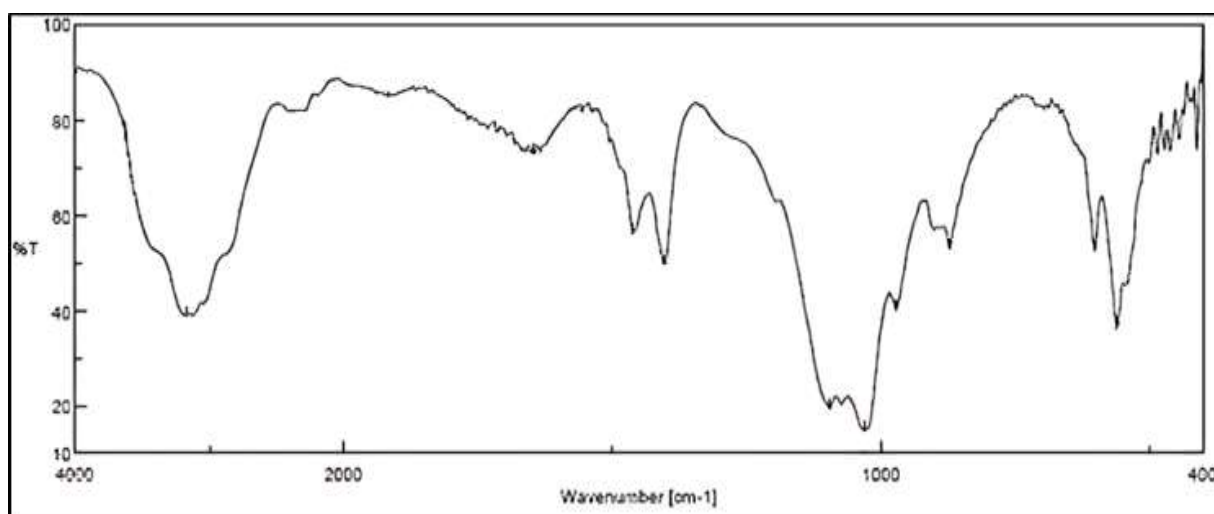


Fig. 4. FTIR spectrum of MAP fertilizer obtained from lignocellulosic biomass.

The thermogravimetric analysis (TGA) study of the obtained lignocellulosic biomass-based magnesium ammonium phosphate fertilizer is given in Fig. 5, gives information about its thermal stability and decomposition reaction during programmed heating. The TGA curves and their corresponding DTG second derivatives showed weight-loss stages, each within certain temperature ranges as well as thermal events.

The first weight-loss step observed for the temperature range 25-150°C could be due to evaporation of physically adsorbed moisture and volatile species from the sample. The second degradation step at 60-260 °C correspond to a more significant weight loss that could be attributed to the thermal decomposition of labile organic functionals, such as carboxylate groups and it was also coupled with CO_2 evolution. In the next temperature region 260-700 °C, additional mass loss can be observed that is ascribed to decomposition of more thermally stable organic compounds and functionalities incorporated in the biomass structure. At temperatures above 700°C, the residual mass is representative of inorganic components like metal oxides or ash-forming compounds, which stay behind as non-volatile residues [52-54].

Overall, the TGA results validate the thermal stability of the lignocellulosic biomass-derived MAP fertilizer up to approximately 200°C and suggests its possible use in agricultural systems with moderate thermal conditions. The results provide useful knowledge for the optimization of the processing and application parameters in relation to maintaining the structural and functional activity of fertilizers.

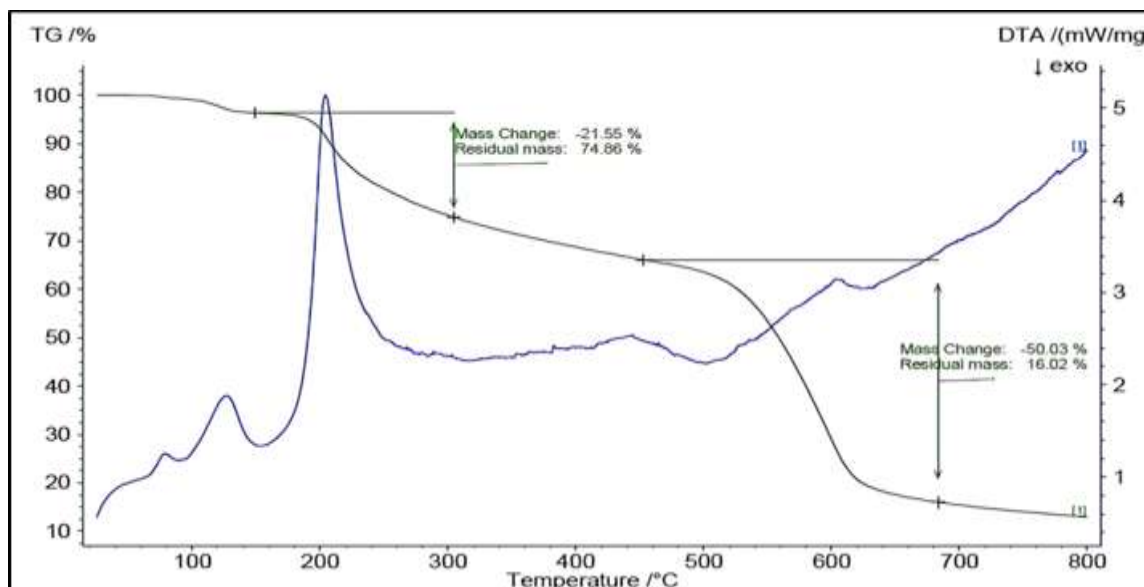


Fig. 5. Thermogravimetric (TGA) of lignocellulosic biomass based magnesium ammonium phosphate (MAP).

Nutrient Release Experiment

The nutrient release characteristics of synthesized material were compared with that of commercial urea and diammonium phosphate (DAP) via leaching column studies. Pretreated and neutralized sand (particle size: 2mm) was packed uniformly in cylindrical glass columns (internal dia. 5cm, length 50 cm). Fertilizer test, equal amounts of nitrogen from the tests were added to separate columns in between two sand layers.

Top outlet of each column received tap water, and leachates were collected at regular periods through the bottom outlet. The nitrogen and phosphorus content of the eluted samples was measured by standard analysis. This system allowed the determination of nutrient release kinetics and a comparison between their performance in sustained-release.

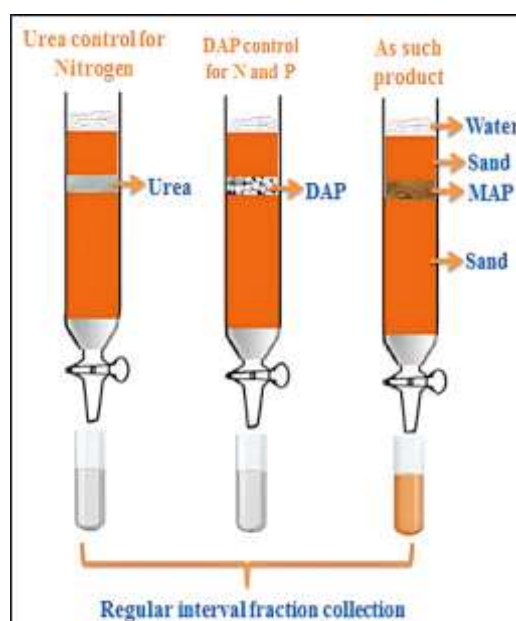


Fig. 6. Dissolution characteristics of magnesium ammonium phosphate (MAP) derived from lignocellulosic biomass and commercial fertilizers.

The available Nitrogen (N) accumulation curves of Fig. 7 indicate different release behaviors among the selected fertilizers. The release pattern of the commercial urea was more rapid, nearly 68 and 96% of the total nitrogen was released in 10 and in 60 h of incubation. On the other hand, diammonium phosphate (DAP) showed a moderate release with 22% and 50% release of N over similar durations indicating a slow-release nature relatively to urea.

The prepared lignocellulosic biomass-derived MAP fertilizer also showed the slowest release rate, in which 7.5% and only 34.8% of N was released after half an hour at 10 h and 60 h, respectively. This sustained release pattern went on constantly and N cumulative release increased up to about 97% at 280 h after incubation. The temporal release profile of the MAP formulation was significantly more linear and extended thanpared to commercial fertilizers. Whereas N-release from urea and DAP reached a plateau at 60 and 130 h, respectively, very little further release took place thereafter.

This means that when lignocellulosic biomass is used as a part of the MAP matrix, it is capable to modify in an efficient way the rate of N release and delay nutrient availability which can minimize leaching losses. The sustained and prolonged release nature of MAP recognises as a potential highly efficient SRF product for enhanced N fertilizer-use efficiency in agricultural practice.

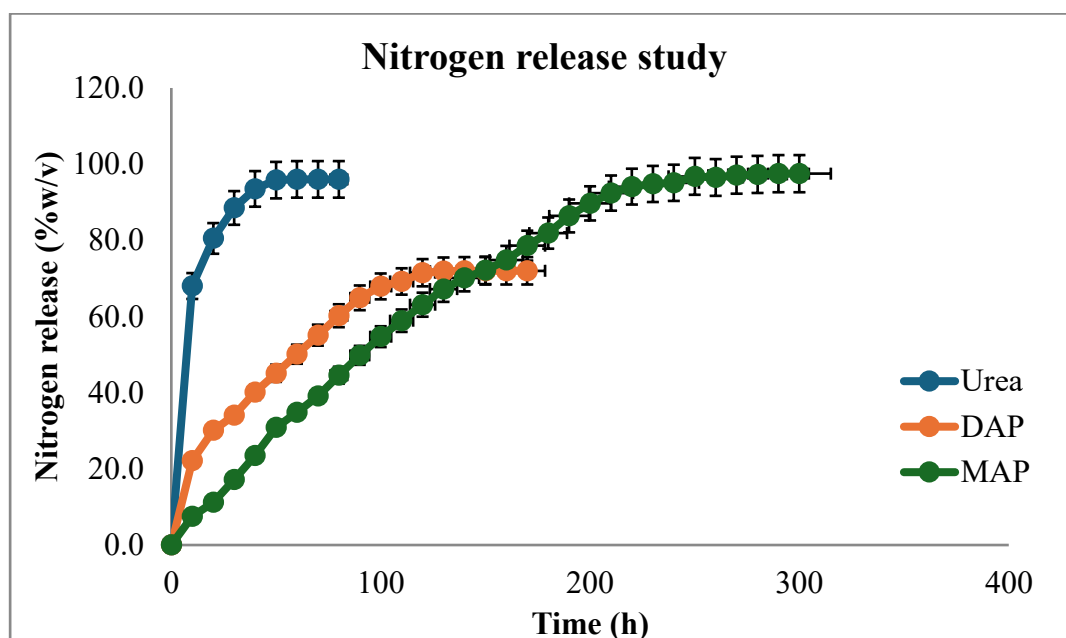


Fig. 7. Nitrogen release kinetics of synthesized lignocellulosic biomass derived magnesium ammonium phosphate (MAP) vs commercial Urea and DAP fertilizers

The total P release profiles of the two biochars are presented in Fig. 8. Investigation was carried out with the native and manufactured lignocellulosic biomass-derived magnesium ammonium phosphate (MAP) fertilizer using commercial diammonium phosphate (DAP). In soil, plant-available P is controlled by precipitation dissolution equilibria, adsorption and desorption at clay and oxide surfaces, and biological immobilization or mineralization mechanisms that together define the release curves. Fig. 8. Nitrogen release profile of lignocellulosic biomass derived synthetic magnesium ammonium phosphate (MAP) in comparison with commercial Urea and DAP fertilizers

The total phosphorus (P) release behavior is depicted in Fig. 9. Commercial DAP and synthesized lignocellulosic biomass-based magnesium ammonium phosphate (MAP) fertilizers were used. Plant-available P in soils is controlled by precipitation dissolution equilibria, adsorption and desorption on clay and oxide surfaces, biological immobilization or mineralization, and this process combination determines the form of resulting release curves.

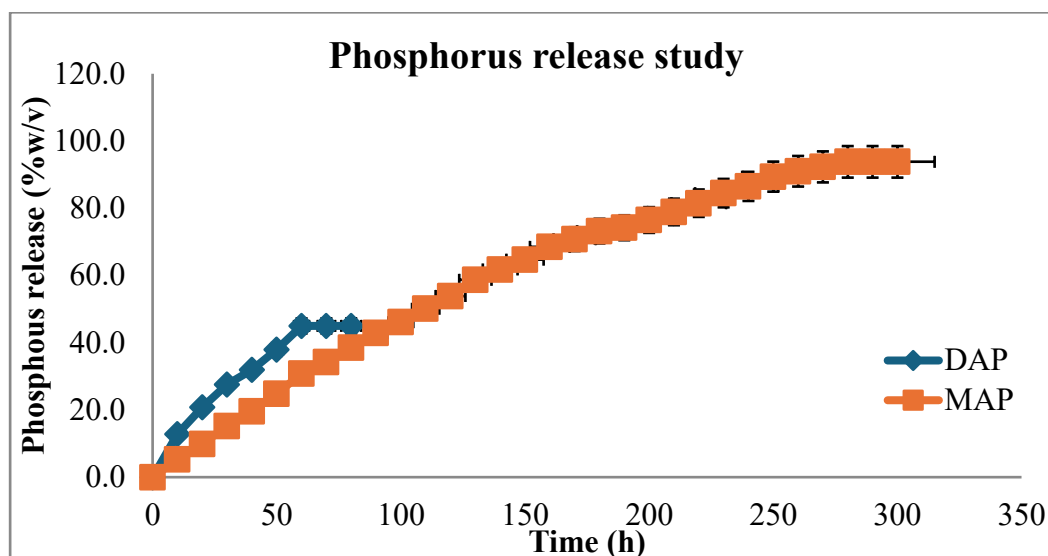


Fig. 8. Phosphorous release pattern of laboratory synthesized lignocellulosic biomass based MAP with that of commercial DAP fertilizer

DAP effluxed rapidly initially, but was due to plateauing at around 45% of its total P content, with ~55% not yet leached from the column. This partial recovery is consistent with the finding that phosphate anions strongly adsorb to bentonite and related aluminosilicate clays via surface complexation and interlayer adsorption, but are also temporarily sequestered by soil microorganisms [55,56].

On the contrary, the release of lignocellulosic biomass based magnesium ammonium phosphate (MAP) fertilizer was significantly slower and nearly linear. Residual loss of P was small during initial weeks, and cumulative liberation pattern was linear all along the 280h test and exceeded DAP level at last. Attenuated kinetics are due to (i) slow dissolution of the LCB derived MAP phase, (ii) diffusional resistance due to the LCB matrix in which mixtures are entwined and progressive degradation leads to stepwise exposure of additional fertilizer surfaces, and (iii) local pH and microbial effects connected with the biomass constituent [57]. These findings indicate that the incorporation of MAP into a lignocellulosic matrix can adjust P availability resulting in controlled nutrient-release as well as reduction of the early-stage losses observed for regular DAP.

Comparative analysis of biochar and other lignocellulosic biomass fertilizers:

Biochar is created through the thermal decomposition of biomasses, such as crop residues, wood and manure at high temperatures to produce carbon rich-porous material with a high surface area and considerable chemical stability in soil environments [58-61]. In spite of being biochar advantageous, that is enhanced nutrient retention and soil amendment potential it has also several environmental and agronomic drawbacks. Pyrolysis can give rise to hazardous organic compounds such as PAHs, dioxins, and furans that would pollute soils and threaten ecosystems and human health [30,31]. In addition, biochar has the potential to adsorb or desorb heavy metals, such as Cu, Cd, Cr, Ni and Zn and their mobility and phytoavailability is strongly controlled by soil pH [24,26,27]. In specific environments, such as acidic or basic pH, the mobility of these metals can be enhanced and affect plant-related health and food safety. While initially biochar can adsorb and immobilize heavy metals, this property loses effect over time because of ageing, oxidation or microbial decomposition (Lu et al., 2013). Furthermore, the bioavailability of native organic pollutants such as phenols could increase after biochar amendment due to promotion of microbial activity, which in turn may stimulate degradation and create an additional level of risk profile [32,33].

Lignocellulosic biomass derived fertilizers on the other hand, present a more environmentally friendly and sustainable option. Mechanisms for their operation include action similar to that described here for SRF's, as well as fertilizers based upon plant-derived renewable feedstocks processed under lower temperature or chemical conditions without the production of hazardous waste by-products associated with high temperature pyrolysis. They serve as biodegradable nutrient carriers, for controlled slow release of nutrients in synchrony with plant uptake dynamics to reduce losses of nutrients due to leaching, and surface runoff [43, 44, 51]. The formulations provide reduced chances of introduction of heavy metals or persistent organic pollutants into soil and have small effects on the pH of the soil. Other benefits are improved soil fertility, reduced dependency on chemical

fertilizer, optimal exploitation of agricultural residues, economy in relation to no-tillage systems under single-dose application strategies and higher cost-effectiveness, compatibility with single-dose applications management strategy and higher credibility. In general, lignocellulosic biomass-derived fertilizers provide multi-nutrient delivery systems necessary for sustainable agronomic practices that promote soil health, nutrient-use efficiency and environmental quality [25,34–35].

Evaluation of lignocellulosic biomass based magnesium ammonium (MAP) phosphate fertilizer use in the tomato crop (*Solanum lycopersicum* L.):

The magnesium ammonium phosphate (MAP) fertilizer prepared from synthesized lignocellulosic biomass and commercially available nitrate-based urea, diammonium phosphate (DAP), were used as sources of nitrogen and phosphorus in the field trials. The field trial was aimed to assess the nutrient release pattern, nutrient utilization efficacy and agronomic performance of lignocellulosic biomass derived MAP fertilizer compared with standard fertilizers used in practice. Potassium was applied in the form of muriate of potash (MOP) from a commercial source.

The experimental field was well prepared using standard agronomic practices before transplanting. Fertilizer was incorporated one time as a basal at transplanting. The fertilizer was applied at a constant rate of 1:0.5:0.5 kg N:P₂O₅:K₂O per m² in all treatments. Fertilized are under root zone at 3–4 cm distance from the base of the transplant to have efficient nutrient availability and uptake. No top-dressing fertilizer was applied during the cropping season to evaluate, soil nutrient supplying available and sustained release characteristics and nutrient utilization efficiency of the lignocellulosic biomass based MAP fertilizer in comparison with standard fertilizers.

Control treatments:

A non-chemical fertilization control treatment with natural soil N availability functioned as the control, and was used for that assessment. A control treatment receiving the normal recommended dose of fertilizer (RDF)

as 100 per cent applied in the form of urea, Diammonium phosphate, and muriate of potash was included. These control treatments offered a yardstick against which the agronomic efficiency and nutrient use efficiency of developed or slow release fertilizer formulations could be evaluated.

Irrigation and Monitoring

All treatments are uniformly irrigated to eliminate moisture induced variation. Treatment effects on growth parameters, yield attributes, nutrient uptake and soil fertility are regularly recorded.

Field trial plant images:

Individual plant physical appearance



Fig. 9. Unfertilized, commercial and lignocellulosic biomass magnesium ammonium phosphate (MAP) fertilized Individual plant appearance

Set of plants physical appearance



Fig. 10. Untreated, commercial and lignocellulosic biomass (LB) magnesium ammonium phosphate (MAP) treated plants physical form

Graph of data and observed:

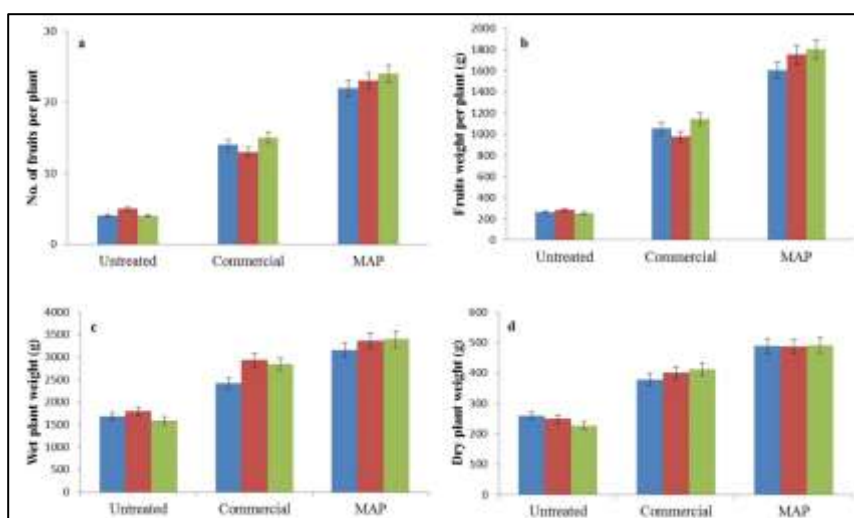


Fig. 10. Profile of No. of fruits plant 1 (a) , Fruits weight plant 1 (b) , wet plant weight (c) and dry plant weight (d) treatment between untreated, commercial Fertilizers and Lignocelolic biomass based magnesium ammonium phosphate treatments (MAP).

The synthesized lignocellulosic biomass derived magnesium ammonium phosphate (MAP) Fertilizer were tested on tomato (*Solanum lycopersicum* L.) with their compatible control treatments. Visual differences in plant vigor were apparent between treatments during the field trial. Plants supplied with the lignocellulosic biomass derived MAP were far better protected and had higher growth vigour and overall health than those treated with commercial fertilizers, whereas in its absence (i.e. without fertilizer) plants performed worst.

Leaf chlorosis was evident from around 40 dp transplant in both the untreated control and CF-treated plants, reflecting a condition of nutrient deficiency or exhaustion. In comparison, dark green and healthy-looking foliage of plants treated with the plant material MAP remained during the entire growing period indicating that nutrient status was maintained over time and that nitrogen nutrition improved.

Although all treatments provided the same amounts of nutrients, commercial fertilized plants showed signs of nutrient depletion sooner and more need for supplementation than MAP-treated plants. The observation revealed that the lignocellulosic biomass based MAP fertilizer made nutrients available to barley crop in a more sustained and regulated manner than other control fertilizers, thus providing continuous nutrient feeding to the crop with better use efficiency of the available nutrients.

Parameter	Untreated	Commercial	MAP
No. of fruits per plant	4.3	14.0	23.0
Fruits weight per plant (g)	268.0	1058.0	1719.7
Wet plant weight (g)	1692.7	2737.3	3313.7
Dry plant weight (g)	246.0	397.7	489.3

Table-2: Average of no. of Fruits/plant (a), Weight of fruits per plant (b), wet wt./plant (c) and dry wt./plant (d) comparison profile between untreated, commercial or lignocellulosic biomass based magnesium ammonium phosphate on Pearl Millet.

Response of Tomato Yield and Biomass Traits to Fertilization:

The response of yield contributing and biomass accumulations to various fertilizer treatments in tomato is shown in table above. The untreated control, commercial fertilizer and lignocellulosic biomass based magnesium ammonium phosphate (MAP) fertilizer treatments showed marked difference.

Fruit yield per plant was considerably different by application of fertilizer. The untreated control had the fewest fruits per plant (4.3), suggesting extreme nutrient starvation. Fruit number was increased to 14.0 per plant in the commercial fertilizer application, and MAP had significantly higher fruit number (23.0 per plant); these number indicated that it increased by about 64% compared with those of the commercial fertilizer treated plants at harvest stage. This increment indicates better nutrient availability and gradual nutrient release when MAP was applied to promote prolonged flowering and fruit setting.

The same trend was also found for fruit weight per plant. The untreated control gave an average of 268.0 g fruit yield per plant as against 1058.0 g recorded for commercial fertilizer treatment. Additionally, the maximum fruit weight (1719.7 g plant⁻¹) produced by MAP was indicative of increased production and transportation of assimilates to developing fruits. Under MAP application, the increased fruit yield might be associated with enhanced N and P use efficiency and decreased nutrient loss.

Fertilizer treatments also had highly significant effect on biomass yield. Wet plant weight increased from 1692.9 g in the untreated control to 2737.3 g with commercial fertilizers, and to 3313.7 g with MAP fertilizer application. Similarly, dry plant weight was reduced by the same degree to 246.0 g, 397.7 g and 489.3 g under untreated, commercial and MAP treatments of irradiated plants. The increased amount of dry matter obtained with MAP application suggests higher photosynthetic capacity and continual nutrient supply during the growing season.

All-in-all that data very effectively display the fact that lignocellulosic-based MAP fertilizer is superior compared to traditional fertilizers in strengthening the tomato yield attributes and biomass, underlining its potential uses for optimal nutrient sources.

CONCLUSION

The current study shows that lignocellulosic biomass-derived MAP has great potential as an efficient slow-release N and P source, surpassing traditional urea and DAP for the timing of nutrient release in accordance with crop needs. Application of

MAP into tomato (*Solanum lycopersicum* L.) increased plant vigor, delayed leaf senescence, and stimulated yield and biomass accumulation up to 64% higher than that obtained with conventional fertilization and by 475% compared with unfertilized plants. The findings of this study demonstrate the ability of biomass derived MAP to enhance nutrient-use efficiency, improve crop productivity, mitigate nutrient loss and reduce environmental pollution.

Taken together, lignocellulosic biomass derived MAP fertiliser is an environmentally friendly and sustainable substitute of mineral fertilisers for intensive horticulture. Future research needs to support RCF by conducting long-term, multi-location investigations and verifying its agronomic performance, economic feasibility as well as scalability under different soil types and climatic zones. These efforts will provide a more comprehensive insight into its prospective role in sustainable and circular agriculture.

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Conflicts of interest

Notes and references the authors have no competing financial interest.

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